

Why are we reinventing basic geometry?

Rearranging areas and dot products are valid methods, but to me, they always felt more consequential than fundamental.

Now to the thought experiment, say I texted you that I walked straight from point A to B. Would you believe it? If I sent you pictures of me at A and B? I could have taken a detour, so no. Hence, sufficient evidence requires a video of me walking straight from A to B.

A video is made of frames, how do we translate those frames into math?

Suppose the smallest discrete unit of length be ϵ , the frame rate of the video would hence be $1/\epsilon$. Would $\epsilon + \epsilon + \epsilon + \dots$ convey information that I have walked say 5 cm? No, it contains no positional data, it is just a collection of random points. $\epsilon + 2\epsilon + 3\epsilon + 4\epsilon + \dots + 4.99999\epsilon + 5\epsilon$ would however, contain enough information that I have walked 5 cm.

Why must it be an arithmetic series?

Additive maps: $f(m+n) = f(m) + f(n)$ since line is uniform.

Naturally, that would result in a **squared** relationship between the **magnitude** of a line and its "**relative history**".

Relative history can be seen as the file size of the video.

Now, we take for granted that humans can experience time, however, pure numerical data cannot choose forwards or backwards. It is indistinguishable whether I went from A to B or B to A. The solution is displacement, if we started at the midpoint between A and B, and expanded outwards in both directions, there would be no contradictions, just pure displacement. Sadly, it also means that we need to forgo single directions. The relative history will now look like $2 * (\epsilon + 2\epsilon + 3\epsilon + 4\epsilon + \dots + 2.499999\epsilon + 2.5\epsilon)$. The midpoint expansion is useful when we need to combine different vectors. In order to form a direction, you need at minimum two ϵ points, we can call those two points the "past" and the "future". When we try to combine different vectors, there will be loss of information, for example, we cannot restore our original vectors from our resultant vector. What if, instead of forcing ourselves down one path, we stay where the vectors intersect, that is, in the present, where we can see all possible pasts and futures without losing any information. The Origin is the perfect intersection for all midpoints/presents since it is zero.

If we were to add two vectors together, it would look like "**X**", with 4 initial directions, intersecting midway at the origin.

How do we translate the previous intuition into concrete steps?

Notice that each direction can mix with 2 other directions, so the total number of resultant directions will be 2^N for N vectors.

Adding videos together requires overlaying frames, it would look like

$$(2E_a+2E_b) + (4E_a+4E_b) + (6E_a+6E_b) + \dots = 2E_c + 4E_c + 6E_c \dots$$

For simplicity, we only take single frames when the amount of frames used is proportionate to the magnitude of each vector. Eg. if $a=2$, $b=3$, $c=4$

$$(3E_a+4E_a+4E_b+5E_b+6E_b) + (7E_a+8E_a+10E_b+11E_b+12E_b) + \dots =$$

$$1 \text{ to } 4E_c + 9 \text{ to } 12E_c + 17 \text{ to } 20E_c + \dots$$

Notice there are missing frames from positive A and B, this is because the other half is being used to form other resultants. Within each bracket, multiple frames are overlayed over each other, which leads to information loss since they are compressed into a resultant direction. We do **not** sequence frames because direction is formed between 2 Epsilon points, sequencing frames would **not** form the resultant direction. (See $P_i=4$ misnomer)

In order to combat information loss in one resultant frame, there will be corresponding information in other respective resultant frames such that they will reinforce each another. (similar to how dot products cancel out)

$$0.5(A+B)+0.5(A-B) = A. \quad 0.5(A+B)+0.5(B-A) = B. \quad A \text{ and } B \text{ are recovered.}$$

This also forces equal “frame sharing” so each resultant needs have the same frame rate.

The resultant vector C has $1/2E$ frame rate. Since data must be conserved, $(E_a+E_b) + (3E_a+3E_b) + (5E_a+5E_b)\dots = E_d + 3E_d + 5E_d \dots$ The resultant vector D frame rate is also halved. The frame rate being halved is a consequence of frames being split/overlayed amongst all possible resultant vectors.

The "relative history" or file size has not changed because information is not lost. That is why, by comparing relative proportions:

$$0.5A^2 + 0.5B^2 = 0.25C^2 + 0.25D^2. \quad \text{That is, the parallelogram law, when } C=D, \text{ it would be the Pythagorean theorem.}$$

If we were to combine N vectors, the frame splitting will get exponential, 2^N because that is how many possible choices of past and futures there are, which would be a Rademacher–Parseval identity, where all cross-terms cancel out.